GALGOTIAS UNIVERSITY

ADMISSION NUMBER
[redacted]

School of Computer Applications and Technology

Semester End Examination - Jan 2026

Masters of Computer Applications

Duration : 120 Minutes
Max Marks : 50

Sem I - E1PA107T - Digital Computer Organization

General Instructions
Answer to the specific question asked
Draw neat, labelled diagrams wherever necessary
Approved data hand books are allowed subject to verification by the Invigilator

1)	Define general-purpose registers and list their advantages.	CO1(5)
2)	Decribe the sub-cycles of instruction execution such as fetch, decode, and execute, with an example.	CO2(5)
3)	Explain how cache memory improves CPU performance with the help of an example.	CO3(5)
4)	Design a 4-bit shift register capable of left and right shift operations. Draw the circuit diagram, explain its operation, and write the register transfer micro-operations for shifting.	CO1(7)
5)	Analyze the objectives of parallel processing in a digital system. Explain in detail how it enhances CPU performance, throughput, and efficiency, and discuss potential trade-offs or limitations in implementing parallel architectures.	CO2(7)
6)	Analyze the concept of cache memory. Explain cache design issues and evaluate cache performance using address mapping techniques and replacement policies. (	CO3(7)
7)	A processor intended for scientific computation must handle large signed multiplications efficiently. Evaluate different multiplication techniques and justify whether Booth's algorithm or an array multiplier is more suitable for this application.	CO2(14)
	OR	
	Evaluate different types of ROM memories. Critically analyze their role in computer systems and justify their use compared to RAM in terms of functionality and reliability.	CO3(14)